CHAPTER 41
Functions of Several Variables

MULTIVARIATE FUNCTIONS AND THEIR GRAPHS

41.1 Sketch the cylinder $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

▮ See Fig. 41-1. The surface is generated by taking the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ in the xy-plane and moving it parallel to the z-axis (z is the missing variable).

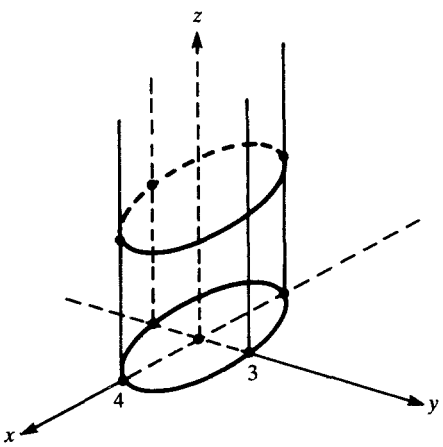

Fig. 41-1

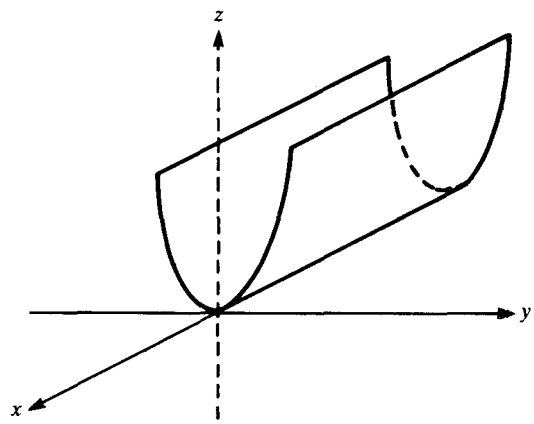

Fig. 41-2

41.2 Describe and sketch the cylinder $z = y^2$.

▮ See Fig. 41-2. Take the parabola in the yz-plane $z = y^2$ and move it parallel to the x-axis (x is the missing variable).

41.3 Describe and sketch the graph of $2x + 3y = 6$.

▮ See Fig. 41-3. The graph is a plane, obtained by taking the line $2x + 3y = 6$, lying in the xy-plane, and moving it parallel to the z-axis.

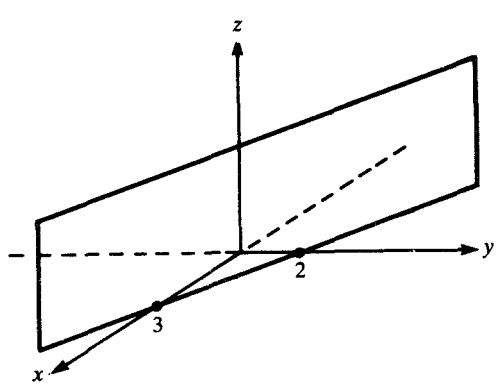

Fig. 41-3

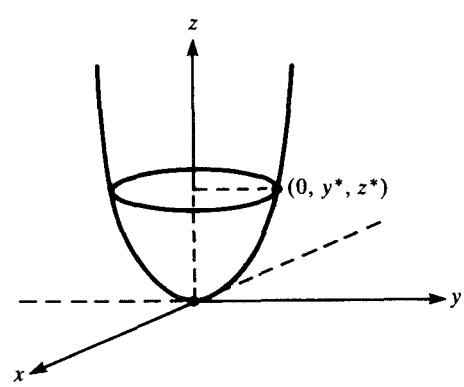

Fig. 41-4

41.4 Write the equation for the surface obtained by revolving the curve $z = y^2$ (in the yz-plane) about the z-axis.

▮ When the point $(0, y^*, z^*)$ on $z = y^2$ is rotated about the z-axis (see Fig. 41-4), consider any resulting point (x, y, z). Clearly, $z = z^*$ and $\sqrt{x^2 + y^2} = y^*$. Hence, $x^2 + y^2 = (y^*)^2 = z^* = z$. So, the resulting points satisfy the equation $z = x^2 + y^2$.

41.5 Write an equation for the surface obtained by rotating a curve $f(y, z) = 0$ (in the yz-plane) about the z-axis.

▮ This is a generalization of Problem 41.4. A point $(0, y^*, z^*)$ on the curve yields points (x, y, z), where $z = z^*$ and $\sqrt{x^2 + y^2} = y^*$. Hence, the point (x, y, z) satisfies the equation $f(\sqrt{x^2 + y^2}, z) = 0$.

41.6 Write an equation for the surface obtained by rotating the curve $\frac{y^2}{4} + \frac{z^2}{25} = 1$ (in the yz-plane) about the z-axis.

▮ By Problem 41.5, the equation is obtained by replacing y by $\sqrt{x^2 + y^2}$ in the original equation. So, we get

$$\frac{(\sqrt{x^2 + y^2})^2}{4} + \frac{z^2}{25} = 1 \qquad \frac{x^2 + y^2}{4} + \frac{z^2}{25} = 1 \qquad \frac{x^2}{4} + \frac{y^2}{4} + \frac{z^2}{25} = 1$$

an ellipsoid.

41.7 Write an equation of the surface obtained by rotating the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ (in the xy-plane) about the x-axis.

▮ By analogy with Problem 41.5, we replace y by $\sqrt{y^2 + z^2}$, obtaining $\frac{x^2}{9} - \frac{y^2 + z^2}{4} = 1$, $\frac{x^2}{9} - \frac{y^2}{4} - \frac{z^2}{4} = 1$. (This surface is called a hyperboloid of two sheets.)

41.8 Write an equation of the surface obtained by rotating the line $z = 2y$ (in the yz-plane) about the z-axis.

▮ By Problem 41.5, an equation is $z = 2\sqrt{x^2 + y^2}$, $z^2 = 4(x^2 + y^2)$. This is a cone (with both nappes) having the z-axis as axis of symmetry (see Fig. 41-5).

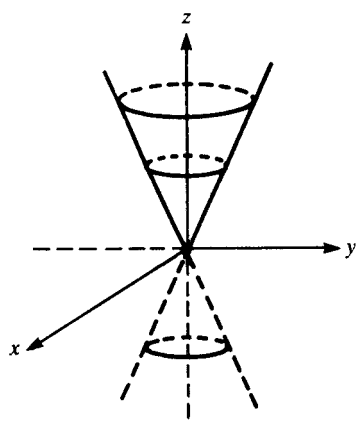

Fig. 41-5

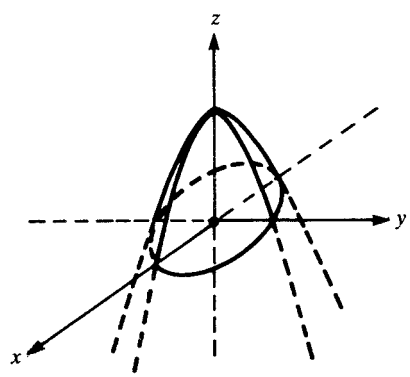

Fig. 41-6

41.9 Write an equation of the surface obtained by rotating the parabola $z = 4 - x^2$ (in the xz-plane) about the z-axis. (See Fig. 41-6).

▮ By Problem 41.5, an equation is $z = 4 - (\sqrt{x^2 + y^2})^2$, $z = 4 - (x^2 + y^2)$. This is a circular paraboloid.

41.10 Describe and sketch the surface obtained by rotating the curve $z = |y|$ (in the yz-plane) about the z-axis.

▮ By Problem 41.5, an equation is $z = |\sqrt{x^2 + y^2}|$, which is equivalent to $z^2 = x^2 + y^2$ for $z \geq 0$. This is a right circular cone (with a 90° apex angle); see Fig. 41-7.

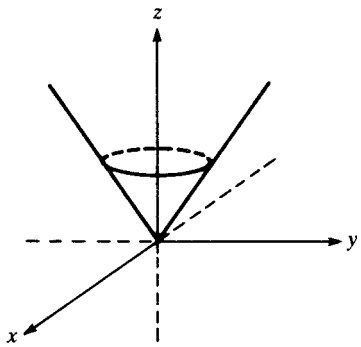

Fig. 41-7

41.11 Of what curve in the yz-plane is the surface $x^2 + y^2 - z^2 = 1$ a surface of revolution?

▮ It is equivalent to $(\sqrt{x^2 + y^2})^2 - z^2 = 1$, which is obtained from the hyperbola $y^2 - z^2 = 1$ by rotation around the z-axis.

41.12 From what curve in the xy-plane is the surface $9x^2 + 25y^2 + 9z^2 = 225$ obtained by rotation about the y-axis?

▮ It is equivalent to $9(\sqrt{x^2 + z^2})^2 + 25y^2 = 225$, which is obtained from $9x^2 + 25y^2 = 225$ by rotation about the y-axis. The latter curve is the ellipse $\dfrac{x^2}{25} + \dfrac{y^2}{9} = 1$.

41.13 Describe and sketch the graph of $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$, where $a, b, c > 0$.

▮ See Fig. 41-8. Each section parallel to one of the coordinate planes is an ellipse (or a point or nothing). The surface is bounded, $|x| \le a$, $|y| \le b$, $|z| \le c$. This surface is called an ellipsoid. When $a = b = c$, the surface is a sphere.

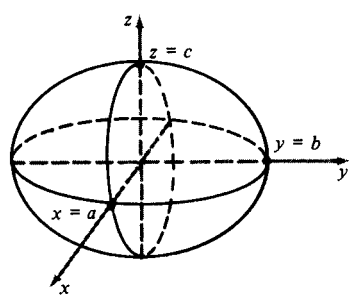

Fig. 41-8

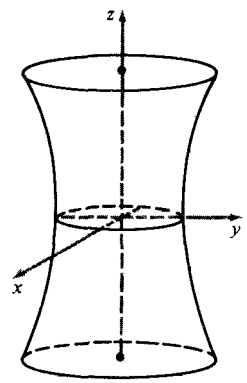

Fig. 41-9

41.14 Describe and sketch the graph of $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$, where $a, b, c > 0$.

▮ See Fig. 41-9. Each section $z = k$ parallel to the xy-plane cuts out an ellipse. The ellipses get bigger as $|z|$ increases. (For $z = 0$, the xy-plane, the section is the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$.) For sections made by planes $x = k$, we obtain hyperbolas, and, similarly for sections made by planes $y = k$. The surface is called a hyperboloid of one sheet.

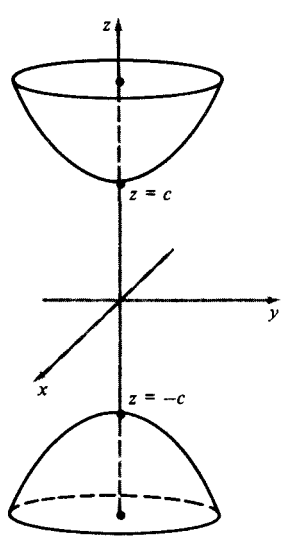

Fig. 41-10